



PH142 Review: Week 1

PH142 GSI Team
Summer 2026



Announcements

- **Quiz04:** due 7/9 at 11:59pm
- **Quiz05:** due 7/10 at 11:59pm
 - Daily quizzes are available all day (24h)
 - Once opened, you have 30 minutes to complete the quiz



Learning Objectives

- Key technologies & Resources
- Reviews the core content from lectures 1-3
 - PPDAC
 - Variable types
 - Relationship between 2 variables
 - Intro to R
 - Data visualization



Key Technologies & Resources

bCourses:

Week 1: Introduction and working with data

 [Learning Objectives](#)

 [Online Study Materials](#)

 [Class Time](#)

 [Assignments](#)

 [Additional Resources](#)

Learning Objectives

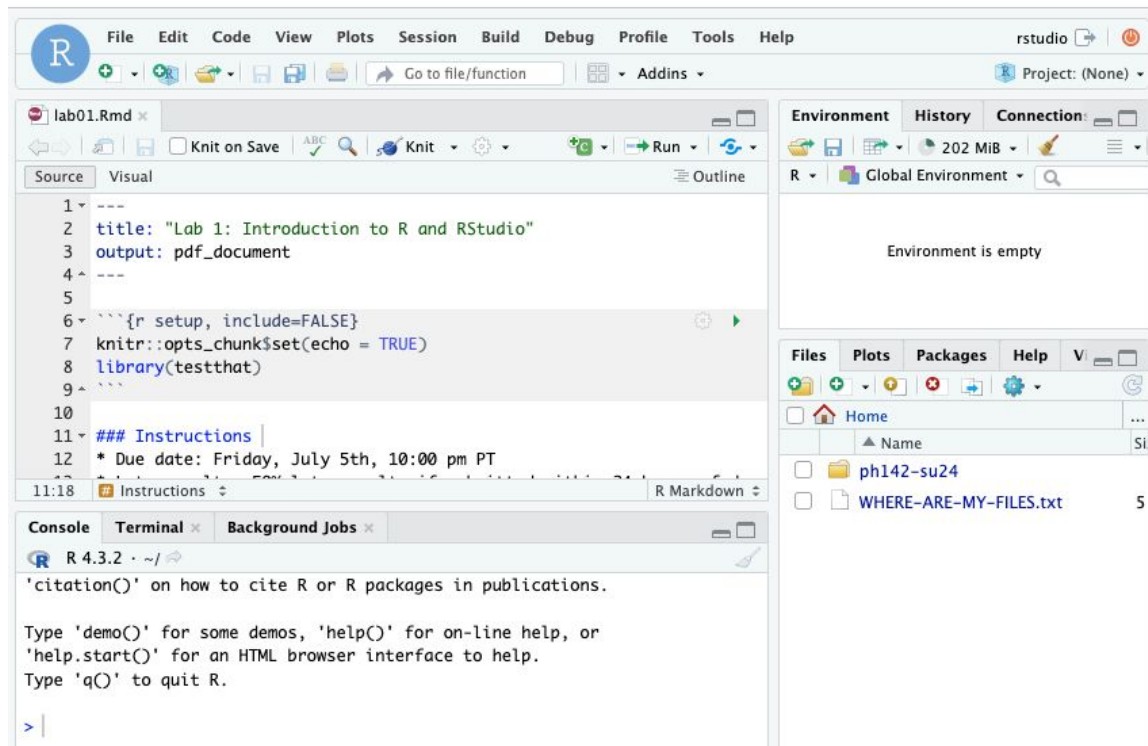
This week, you will learn to:

- Access R and R studio through the datahub
- Open and explore a dataset
- Create some visual and numeric summaries
- Estimate a line of best fit using linear regression



Key Technologies & Resources

DataHub:



Key Technologies & Resources

Ed:

ed PBHLTH 142 LEC 001 – Ed Discussion

New Thread

Search

Filter

CATEGORIES

General

Lectures

Labs

Problem Sets

Data project

Datahub-technical

Lab sections

General Mi-Suk Kang Dufour STAFF 11h

Welcome!

General Mi-Suk Kang Dufour STAFF 2d 16

This Week

Missed Lab Section Today

General Aryaman Kohli 3h 1

Submission not showing up on Gradescope

Labs Anonymous 4h 3

Lab 105-106 Section Slides (GSI: Tianyue & Syl...

General Sylvia Song STAFF 6h

Submission Problem

General Anonymous 7h 2

Knitr error during lab session

Labs Anonymous 7h 3 (2 new) 1

Gradescope access

General Josue Parra 9h 1

99 others online

Welcome! #1

M

Mi-Suk Kang Dufour STAFF

2 days ago in General

UNPIN

STAR

WATCH

739 VIEWS

16

Hello and welcome to ph142 summer 2024,

Lectures will be a mix of live and pre-recorded material. The live lectures will be held by zoom at [this link](#). They will be recorded and posted subsequently. Each lecture will be associated with a very short quiz on gradescope - which should be completed within 2 days of the lecture. This helps the teaching team check comprehension of the material and gives you a chance to practice answering questions that are similar to the type of question you will see on exams.

This weeks lectures will orient you to the course and to R and R studio - the software that we will use in the course. We will also discuss visualizing and summarizing data. We will have live lectures on Monday and Friday. Lecture materials are posted on the bcourses site <https://bcourses.berkeley.edu/courses/1535789>

Sections will be held live only and will not be recorded. Sections will give you an opportunity to work through the labs with support, to review material, to work on your data projects with support, and to ask questions. In general there will be two labs, a dedicated review session and a data project support session each week in section.

We have a short needs assessment which will help us to organize support for the course and which will count as a participation activity towards your grade. You can find the questionnaire [here](#)

All assignments should be submitted to gradescope. You should have been added to the gradescope site - if you have any issues accessing the site, please post a message

Key Technologies & Resources

Gradescope:



PH 142

Introduction to probability and statistics in biology and public health

Dashboard

Assignments

Roster

Extensions

Course Settings

Instructors

Mi-Suk Kang Dufour

Tianyue Zhou

Kristen Kim

PH 142 | Summer 2024

Course ID: 803249

Description

Edit your course description on the [Course Settings](#) page.

Things To Do

- Review and publish grades 1
- Review and publish grades 1

Active Assignments

Released

Due (PDT) ▾ Subn

Time Limit: 30 Minutes

Quiz 2

JUL 2, 2024 11:00 AM

JUL 5, 2024 10:00 PM

Late Due Date: JUL 5, 2024 11:59 PM

Lab 01

JUL 1, 2024 11:00 AM

JUL 5, 2024 10:00 PM

Late Due Date: JUL 5, 2024 11:59 PM

Time Limit: 30 Minutes

Quiz 1

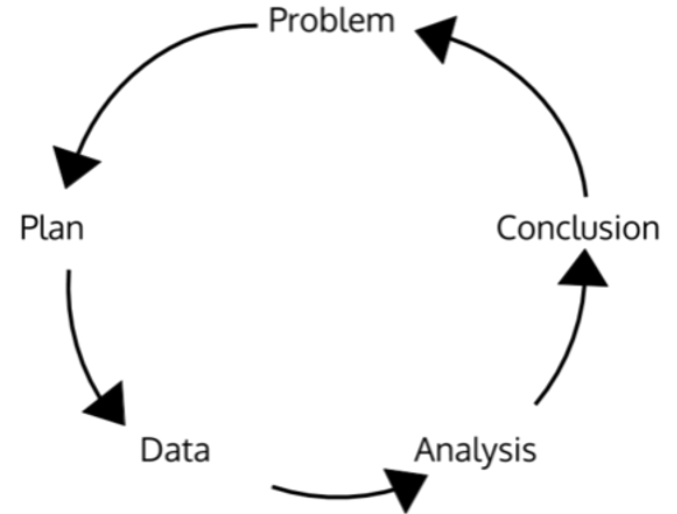
JUL 1, 2024 11:00 AM

JUL 3, 2024 10:00 PM

Late Due Date: JUL 5, 2024 10:00 PM

PPDAC Approach

- **Problem:** A clear statement of what we are trying to achieve
 - problem type – descriptive, predictive, causative/etiologic
- **Plan:** the procedure we use to carry out the study
- **Data:** data collect based on the plan
- **Analysis:** summarization and analysis of the data to answer questions posed by your problem
- **Conclusion:** what you learned from your answers to the problem





Types of Variables

- **Categorical:** a variable that has grouping levels
 - **Nominal:** no underlying order or rank, e.g. blood type, zip code
 - **Ordinal:** with an underlying order or rank, e.g. blood pressure level (low, normal, high)
- **Quantitative:** a numeric variable which you can perform mathematical operations on
 - **Discrete:** can be counted, e.g. the number of cookies in the bag you got from a bakery
 - **Continuous:** can be measured precisely, with a rule or scale, e.g. 5.34 grams of cornstarch



Relationship between 2 variables

Two variables

- X: explanatory variable, independent variable
- Y: response variable, dependent variable
- Scatter plot is helpful to visualize their relationship, and we should pay attention to:
 - The overall pattern
 - Form
 - Strength
 - Any exceptions to the overall pattern
 - Any outliers



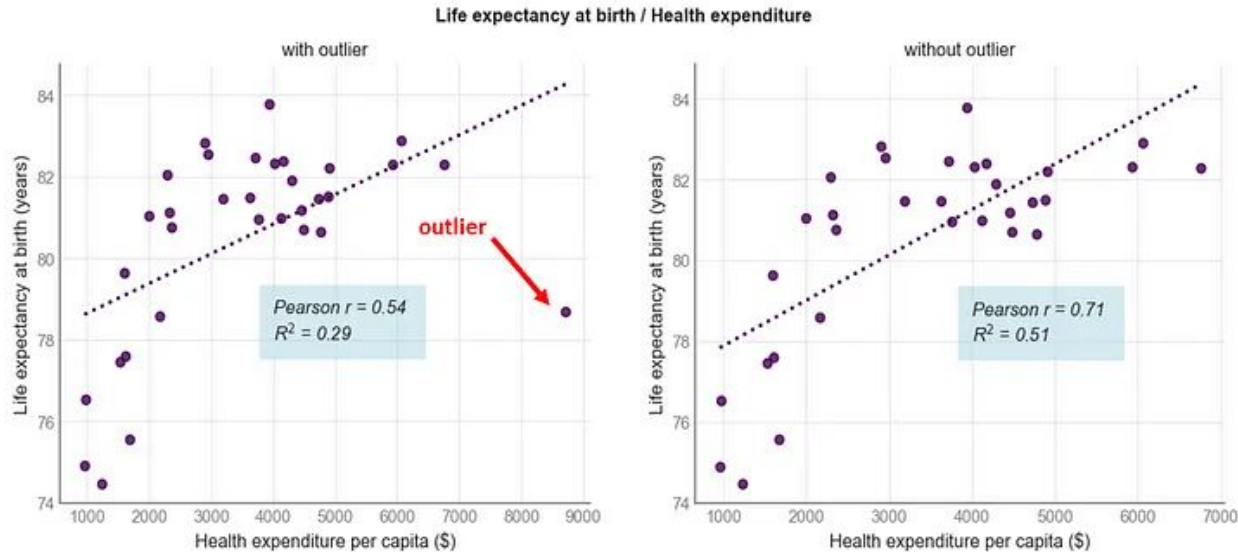
Relationship between 2 variables

Pearson's correlation

- -1 to 1
- 0 indicates no association
- Only useful when the relationship is linear

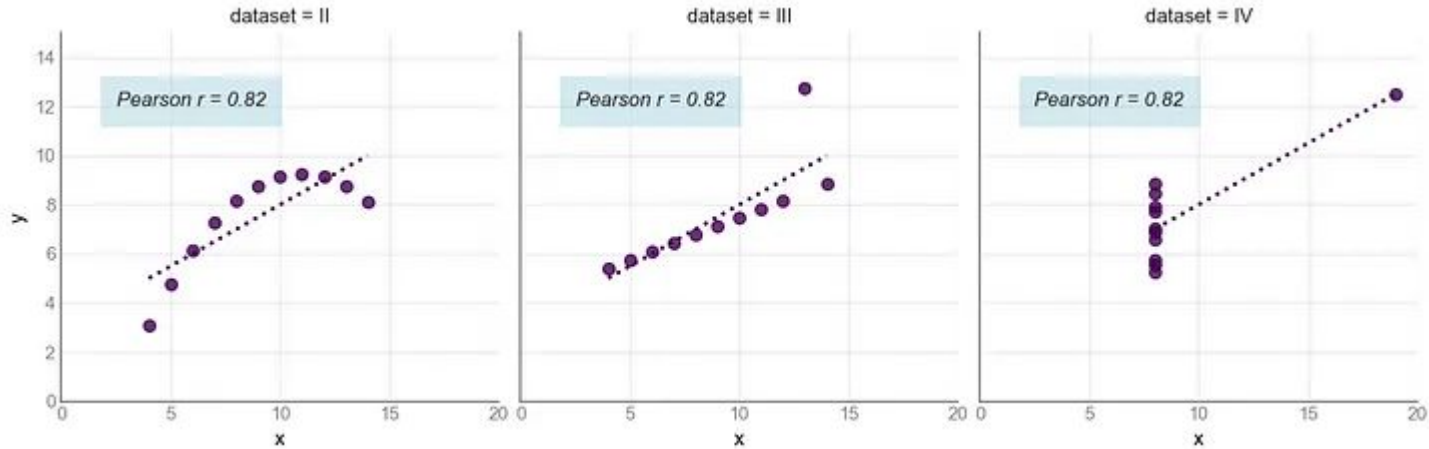
$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Relationship between 2 variables



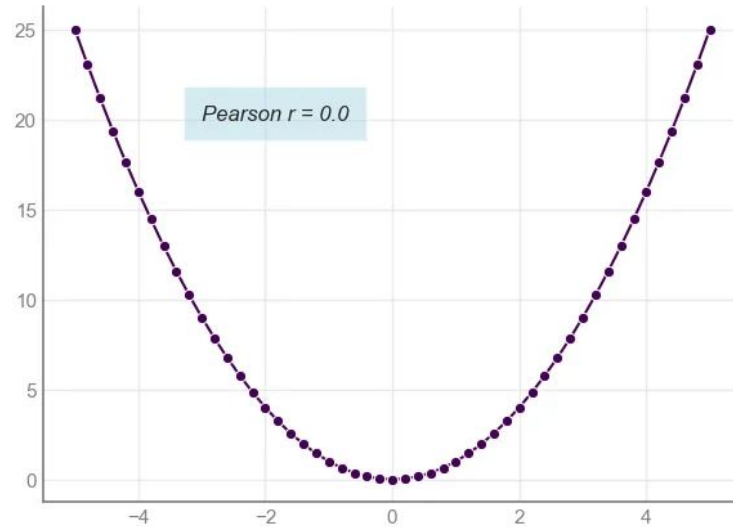
Influence of an outlier

Relationship between 2 variables



High correlation doesn't indicate a linear relationship

Relationship between 2 variables



No correlation doesn't mean no relationship



Intro to R

- Library
 - A library is a package of functions, and you can load this package of functions by running `library(ggplot2)`, `library(dplyr)`
 - Make sure you have them first, otherwise you need to do `install.packages()` first
- Read your data: e.g. how to read a .csv file
 - `library(readr)`
 - `mydata <- read_csv('my_data.csv')`
- Some functions to get a quick look of your data
 - `head(mydata)`: shows the first 6 rows of the dataset
 - `dim(mydata)`: shows the total number of rows by the total number of columns
 - `names(mydata)` or `colnames(mydata)`: shows all the variable names (column names) of the dataset
 - `str(mydata)`: summarizes the information above and more



Data manipulation: functions in library(dplyr)

First, do `library(dplyr)` to have the package in your environment.

- **rename()** → renames variables (columns)
 - `new_dataset <- old_dataset %>% rename(new_name = old_name)`
 - **or:** `new_dataset <- rename(old_dataset, new_name = old_name)`
- **select()** → subsets variables (columns)
 - `smaller_data <- old_data %>% select(variable1, variable2, variable3)`
 - `smaller_data <- select(old_data, variable1, variable2, variable3)`
 - `smaller_data <- select(old_data, variable1:variable3)`
 - **To keep all variables other than variable1:** `smaller_data <- old_data %>% select(- variable1)`
- **arrange()** → orders observations (rows) by a certain variable (column) or variables (columns)
 - `lake_data %>% arrange(ph)`
 - `lake_data %>% arrange(age_data, ph)`



Data manipulation: functions in library(dplyr)

- **filter()** → selects a subset of rows by certain conditions
 - If we want condition A AND condition B to be satisfied, use `,` or `&`
 - If we want condition A OR condition B to be satisfied, use `|` or `%in%`
 - `lake_data %>% filter(age_data == "recent")`
 - `lake_data %>% filter(lakes %in% c("Alligator", "Blue Cypress"))`
 - `lake_data %>% filter(ph > 6 | chlorophyll > 30)`
- **mutate()** → creates new variables
 - `lake_data_new <- lake_data %>% mutate(actual_fish_sample = number_fish_sampled * 100)`
- **group_by()** → groups the data by a categorical variable
 - `lake_data %>% group_by(age_data) %>% summarize(mean_ph = mean(ph))`
- **summarize()** → applies summary functions to calculate statistics
 - `lake_data %>% summarize(mean_ph = mean(ph), sd_ph = sd(ph))`



Visualization of Categorical Data: “ggplot2”

1. Install and load the “ggplot2” package
 - a. `install.packages("ggplot2")`
 - b. `library(ggplot2)`
2. Specify your data, and what to have on the x and y axes
3. Create a plot: `geom_` functions (many options)
 - a. `geom_bar`, `geom_histogram`, `geom_point`, `geom_line`
 - b. Tip: picture how you want to visualize your data in your head first, then pick the function that helps you achieve your goal :)
4. Change the style of your plot
 - a. `labs()` function: update your main title, axis names, caption
 - b. `theme()` function: change the size of your title, font, and position

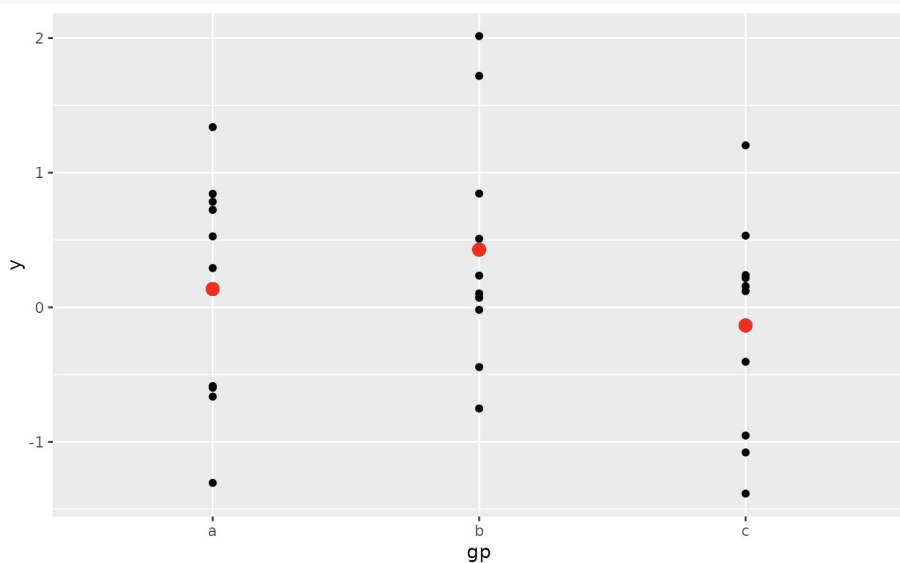
```
ggplot(data = opi_data, aes(x = Mean)) +  
  geom_histogram(col = "white") +  
  labs(x = "Mean opioid prescription rate (per 100 individuals)",  
        y = "Number of states") +  
  theme_minimal(base_size = 15)
```

Visualization of Categorical Data: `geom_point()`

To learn more:

- <https://ggplot2.tidyverse.org/>
- *R for Data Science*:
<https://r4ds.hadley.nz/data-visualize>

```
ggplot(df, aes(gp, y)) +  
  geom_point() +  
  geom_point(data = ds, aes(y = mean), colour = 'red', size = 3)
```



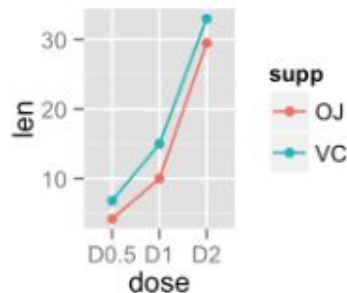
source: <https://ggplot2.tidyverse.org/reference/ggplot.html>

Visualization of Categorical Data: `geom_line()`

To learn more:

- <https://ggplot2.tidyverse.org/>
- *R for Data Science*:
<https://r4ds.hadley.nz/data-visualize>

```
p<-ggplot(df2, aes(x=dose, y=len, group=supp)) +  
  geom_line(aes(color=supp))+  
  geom_point(aes(color=supp))  
p
```

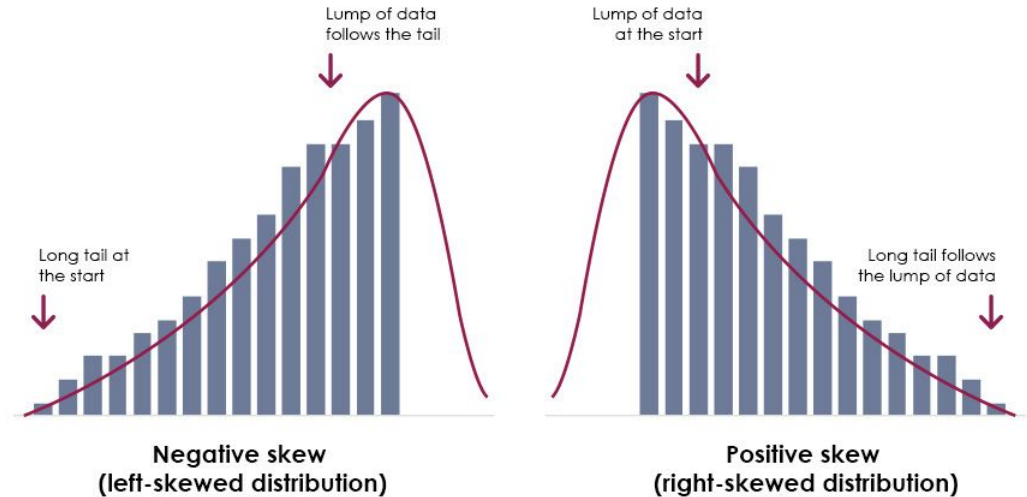


Picture source:

<http://www.sthda.com/english/wiki/ggplot2-line-plot-quick-start-guide-r-software-and-data-visualization>

Measure of Central Tendency

- Mean and median are **approximately equal** when...
 - Distribution is symmetric
 - Data has one peak
 - There are no outliers
- Outliers: large effect on the mean
- Skewed data: mean $\neq$ median
 - Skewed right: mean $>$ median
 - Skewed left: mean $<$ median



Picture source: <https://www.cambridgemaths.org/blogs/skewed-usage-skewed-distribution/>



Measures of Spread

- Range = max - min
- IQR = Q3 - Q1
 - Five number summary in R!

```
CS_dat %>% summarize(min = min(cs_rate),  
  Q1 = quantile(cs_rate, 0.25), median = median(cs_rate),  
  Q3 = quantile(cs_rate, 0.75), max = max(cs_rate))
```

- Sample variance (s^2)
- Sample standard deviation (s)

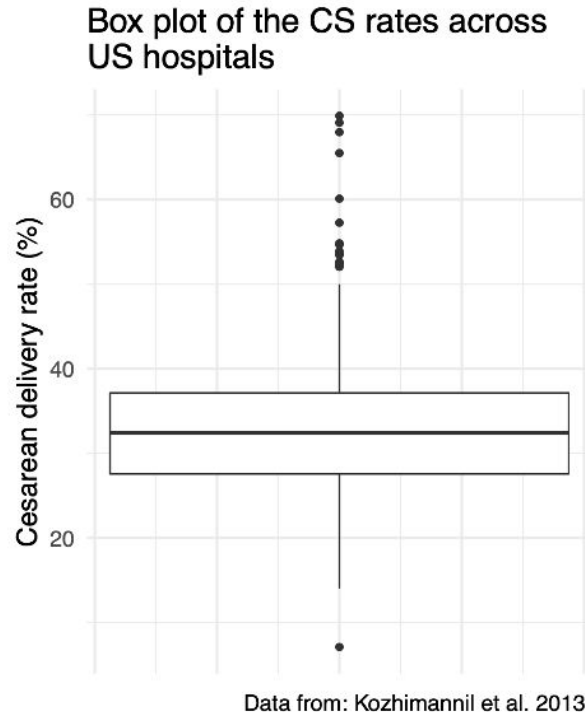
```
CS_dat %>% summarize(cs_sd = sd(cs_rate), cs_var = var(cs_rate))
```

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Box plot

- Center line → median
- Top of box → Q3
- Bottom of box → Q1
- Top of top whisker → max value or highest point that is below $Q3 + 1.5 \times IQR$
- Bottom of bottom whisker → min value or lowest point that is above $Q1 - 1.5 \times IQR$
- Data points above and below whiskers → outliers

```
ggplot(CS_dat, aes(y = cs_rate)) +  
  geom_boxplot() +  
  ylab("Cesarean delivery rate (%)") +  
  labs(title = "Box plot of the CS rates across US hospitals",  
        caption = "Data from: Kozhimannil et al. 2013.") +  
  theme_minimal(base_size = 15) +  
  scale_x_continuous(labels = NULL) # removes the labels from the x axis
```





Common Errors

- Typo ;)
- Used a function from a package that wasn't loaded yet.
- Called a function on an object/variable that has the wrong type.
- Two different code chunks shared the same name.
- The same variable names that are listed in the instructions are used in your work.
- Data/code didn't run – try reloading your previous code chunks first.
- How to examine the output of an object or a line of code in rmd – simply run the code or the name of the object in the code chunk.



See you next week!